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Complete List of Authors:	Lyrio, Julio; Petrobras, EXP/GEOF/MNS Menezes, Paulo T. L.; Universidade do Estado do Rio de Janeiro, DGAP/FGEL/UERJ; Petrobras, EXP/GEOF/MNS Correa, Jorlivan; Petrobras, EXP/GEOF/MNS Viana, Adriano; Petrobras, EXP/GEOF/MNS

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MULTIPHYSICS ANOMALY MAP: A NEW DATA FUSION WORKFLOW FOR GEOPHYSICAL INTERPRETATION

Julio Cesar S. O. Lyrio[1], Paulo T. L. Menezes[1,2], Jorlivan L. Correa[1], and Adriano R. Viana[1]

[1]PETROBRAS–EXP/GEOF/MNS

Av. Henrique Valadares 28, Torre B, 1° andar,

Rio de Janeiro, RJ, Brasil, 20231-030.

[2]DGAP/FGEL/UERJ

Rua São Francisco Xavier 524, 2029a,

Rio de Janeiro, RJ, Brasil, 20550-013.

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ABSTRACT

When collecting and processing geophysical data for exploration, the same geological feature can generate a different response for each rock property being targeted. Typically the units of these responses may differ by several orders of magnitude. Because of that, the combination of geophysical data in integrated interpretation is not a straightforward process and cannot be performed by visual inspection only. The proposed multiphysics anomaly map is a data fusion solution that consists of a spatial representation of the correlation between anomalies detected with different geophysical methods. In the multiphysics anomaly map, we mathematically process geophysical data such as seismic attributes, gravity, magnetic, and resistivity before combining them in a single map. In each data set, anomalous regions of interest, which are problem depend, are selected by the interpreter. Selected anomalies are highlighted through the use of a logistic function, which is specially designed to clip large magnitudes and rescale the range of values, increasing the discrimination of anomalies. The resulting anomalies, named logistic anomalies, represent regions of large probabilities of target-occurrence. This new solution highlights areas where individual interpretations of different geophysical methods correlate, increasing the confidence in the interpretation. We illustrate the effectiveness of our multiphysics anomaly map with the application to real data from onshore and offshore Brazil. In the onshore Recôncavo Basin, the multiphysics anomaly map allows the interpreter to identify a channel where a drilled well found the largest sandstone thickness on the area. In the second example, from offshore Sergipe-Alagoas Basin, the multiphysics anomaly map helps to differentiate between a dry and an oil-bearing channel previously outlined in seismic data. These outcomes, therefore, demonstrate the multiphysics anomaly map is a valid interpretation tool that we believe can be applied to a wide range of geological problems.

INTRODUCTION

In geophysical exploration, the understanding of the subsurface geology in complex regions frequently requires the interpretation of a combination of different geophysical data since some methods may have limited or even no response to a particular rock property. In most cases, the properties recovered from each method may differ by several orders of magnitude from each other, thus making the combination of geophysical data for the purpose of integrated interpretation a difficult process.

Customarily, the interpretation process requires more than ordinary visual inspection of single geophysical maps to adequately reduce uncertainties. Data fusion techniques can simplify such processes by providing a total or partial synthesis of various geophysical quantities.

The use of data combination (or fusion) techniques to support interpretation is not new in geophysics. The geophysical data combination as composite color images is one of the most common methods. According to Duval (1983) this procedure was initially developed by the United States Geological Survey (USGS) to combine aerial gamma-ray data in mineral exploration in South Texas. Composite images using gravity, magnetic and radiometric data were also used by Broome (1990) to assist geological mapping in the Precambrian Canadian Shield. Tilbury and Bush (1991) have also applied image processing techniques coupled with comprehensive analysis of seismic horizon attributes to enhance 3D seismic interpretation in Western Australia.

In seismic interpretation, spectral decomposition analysis of 3D seismic data used in conjunction with RGB (red-green-blue) blending have become a common tool for looking at both structural and stratigraphic features. Examples of such applications can be found in Hall and Trouillot (2004) where RGB and HSB (hue-saturation-brightness) composite images were used in stratigraphy prediction; in Henderson et al. (2007) for whom the use of composite images in spectral decomposition analysis was considered fundamental to obtain information from his innovative volumetric spectral decomposition technique; in McArdle and Ackers (2012) who use it to map thickness variation of thin-beds; and in Cooke et al. (2014) who studied through forward modeling, the color response of High Definition Frequency Decomposition (HDFD) blend around a gas discovery. The spectral decomposition analysis of 3D seismic data, however, can not be considered a typical multiphysics combination tool because it deals with seismic data only.

The development of the data fusion process techniques led to new approaches for geophysical data integration. For instance, Cowan and Cooper (2009) proposed multiplying standardized versions of gravity and magnetic products to construct a map from the Vredefort Dome (South Africa) highlighting areas of positive and negative correlation between both products. Carneiro et al. (2012) have suggested that the analysis of large multivariate datasets could be improved by the use of self-organizing maps (SOM), a technique they applied to integrate geophysical data in a semiautomated geologic mapping of the Brazilian Amazon. A combination of seismic, resistivity and neutron capture data was presented by Altundas and Chugunov (2018) in a multiphysics feasibility modeling workflow for fluid front monitoring that enabled uncertainty analysis and the construction of probability maps of time-lapse changes. Machine learning algorithms exploring the complex relationships among multiple datasets including magnetotelluric, magnetic, electromagnetic, gravity, wells and seismic data, were used by Velasco et al. (2018) to highlight areas of higher CO_2 potential.

We propose a new solution, herein named Multiphysics Anomaly Map (MAM), which results from the fusion of geophysical data from different methods. Following the concepts defined by Cowan and Cooper (2009) we multiply non-linearly normalized versions of different geophysical data to build a spatial representation of the correlation between the anomalous regions detected by each geophysical method, assuming that these anomalies are associated to the same geological source. In our approach, high correlation areas are directly related to the probability of target occurrence. The flexibility of the proposed solution, demonstrated by the application to real data encourages the integration of multiple geophysical methods to create significant and innovative solutions for mineral exploration.

METHODOLOGY

The MAM solution results from a data fusion approach designed to combine data from different geophysical methods such as gravity, magnetic, resistivity, and seismic attributes. The choice of the geophysical methods to be included in the MAM is critical for the methodology success since a single method may have limited or even no response to a particular rock property. Because this choice is problem dependent, it is the role of the interpreter to select among the available methods those whose combined solution provides a total or partial synthesis of the different geophysical data and may be translated into a geological aspect of the problem.

Our methodology essentially consists of combining mathematically highlighted anomalous regions of each geophysical method, which are selected by the interpreter according to each specific geological problem. This combination is performed in such a way that anomalous areas from each method, interpreted as related to the same geological source, will show spatial superposition and consequently will be highlighted. On the other hand, background areas and anomalies related to different sources will be downgraded.

To that end, we introduce a modified logistic function, a mathematical tool used to transform each geophysical attribute to allow their combination. Then, we discuss how geophysical attributes are suited to take maximum advantage of the logistic function characteristics and how these attributes are combined to produce the multiphysics anomaly map.

Logistic function

A logistic function or logistic curve is a common non-linear mathematical function showing a typical S-shape (sigmoid) curve often used in statistics and machine learning techniques. The function was named by Pierre-François Verhulst, a Belgian mathematician who developed it for population growth studies (Verhulst, 1845). In statistics, the logistic function is associated with the logistic distribution, one of the most useful probability distributions, playing an important role during studies of growth models. Logistic functions are often used in artificial neural networks and learning algorithms to introduce nonlinearity into the models. Logistic functions have also been applied to a wide number of areas like physics, chemistry, biology, ecology, economics, geosciences, sociology and political sciences, among others. In our methodology the logistic function is used to keep large magnitude data within bounds while preserving the intermediate values, which are important for data combination.

The generic form of the logistic function σ proposed in this work can be described by the equation:

$$\sigma(x) = \frac{L}{1 + e^{-ks(x-x_0)}}, \tag{1}$$

where e is the natural logarithm base, x_0 is the x-value of the sigmoid inflection point, L is the curve maximum value, k is the steepness of the curve, and s is a scaling factor that fits the data domain into the chosen function domain. The introduction of the scaling factor s into our logistic function yields data with distinct domains to produce similar responses. Figure 1 shows two forms of logistic functions for comparison: the standard sigmoid function (dashed line) with parameters L=1, k=1, s=1 and x_0=0, and the proposed modified logistic function (solid line) using parameters L=2, k=0.25, s=1 and x_0=0.

In general, the curve of a logistic function approaches L as x tends to $+\infty$ and it approaches zero as x tends to $-\infty$. Since the range of a function is said to be the set of all output values (σ-values) satisfying the function, the range of a general logistic function is (0, L). On the other hand, the domain of a function is defined as the set of all x-values corresponding to the function and, therefore, the logistic function domain corresponds to the set of real numbers. In practice, due to the nature of the exponential function e^{-x}, it is often sufficient to compute the logistic function over a small domain of real numbers.

Data preparation

Data combination from different geophysical methods usually requires normalization to deal appropriately with the distinct physical units. Normalizing data by dividing the entire dataset by a chosen constant value is a common practice. However, its linear nature can create large results for bounding values of the dataset, which may lead to undesirable results during anomaly combination. The asymptotic characteristic of the logistic function as x tends to $\pm\infty$ seems more suitable for getting the desired normalization results since even extreme x-values will always result in σ-values bounded by the logistic function range.

In this work, we choose to broaden the role of the logistic function to beyond its use as a simple normalization tool. In fact, the logistic function as implemented here works like a penalty function, where part of the input values will be favored while the remaining part will be penalized. We set such arrangement by defining the curve's maximum value L to 2 so that the function range becomes (0,2) and the sigmoid inflection point at x_0 becomes 1. If parameter x_0 can be thought as a threshold value, then any x-values greater than the threshold will result in σ-values larger than unity, which means being favored. On the other hand, x-values smaller than the threshold will be penalized because the σ-values will be smaller than 1. The advantage of such implementation is evident in Figure 1 where one can compare the result of the proposed logistic function (solid line) with that from the standard sigmoid function (dashed line), where all x-values are penalized since the function output equals 1 only at x=$+\infty$.

Another advantage of the logistic function over the linear normalization resides in the control of the steepness (k parameter) of the S-curve. Values of k parameter smaller than the unity are responsible by smooth transitions between strongly favored and strongly penalized values, while k parameter greater than the unity produces a rapid transition. It is important to emphasize that the larger the transition zone, the greater is the preservation of intermediate anomaly values. Figure 1 illustrates the results of the logistic functions with different steepness values.

The transition zone in the logistic function plays an important role in the MAM construction. Because the multiphysics anomaly map is a spatial representation of the correlation of different methods, it's important to keep in mind that strongly penalized areas from one method may significantly reduce the favored areas in the others. Thus, smooth transitions seem to be a more natural choice for the MAM processing. We have empirically chosen the steepness parameter k=0.25 because it partially preserves the importance of some localized remarkable anomalies while stressing areas where anomalies from all methods superpose.

A direct consequence of using small steepness value for the logistic function is the fact that, due to the smoothness, it is necessary to enlarge the function domain to achieve σ-values above 1.99 and below 0.01. We decide to set the domain of our logistic function to the interval (-24,24), as shown in Figure 1, which will produce the approximate range (0.0049,1.9951).

Once the function domain is defined, each geophysical method involved in the construction of the MAM needs to be appropriately scaled to the (-24,24) interval. As the smooth transition zone is important for anomaly preservation, it is fundamental to minimize the influence of outliers in the data. Such procedure aims to assign the maximum domain to the bulk of the data providing maximum discrimination among the data. Data distribution analyses are necessary in order to perform a careful recognition of the outliers. If the distribution approaches a symmetric distribution, we apply the Two Sigma Rule, an empirical rule stating that, for many reasonably symmetric unimodal distributions, approximately 95% of the population lies within two standard deviations of the mean. The two sigma interval is then set to be our data domain. For asymmetric unimodal distributions, we select the 95% lowest values for right-skewed distributions and the 95% highest values for the left-skewed ones.

The chosen data domain is then shifted and centered at zero, which makes the limits of the new domain proportional to the limits of the logistic function. The dataset is then rescaled to the interval (-24,24) by applying a scale factor s that is the ratio between the domain of the logistic function and the data domain.

The choice of the threshold parameter x_0 is of fundamental importance in the construction of the MAM. Since the logistic function is characterized as a penalty function, it is necessary to define the range of anomalous values that would be penalized or favored in each geophysical method. The choice of x_0 is, indeed, an interpretation task because the selection of significant anomalies in each method is problem dependent. The selection of the threshold x_0 must be made with care since it is critical for the final results. The resulting datasets after applying the logistic function to each anomalous data are named logistic anomalies.

Multiphysics Anomaly Map

The wide variety of options resulting from parameter manipulations makes the logistic function a robust and flexible tool for adjusting the multiple geophysical datasets during their preparation towards the final combination into the multiphysics anomaly map.

The fusion of all logistic anomalies into the MAM is accomplished by multiplying all logistic anomalies. Considering that each logistic anomaly ranges between 0 and 2, the final product will vary between 0 and 2^N, where N is the number of logistic anomalies involved in the calculation. As a result, the MAM highlights regions showing the maximum correlation between major features interpreted from different geophysical methods. In the multiphysics anomaly map, we expect areas showing larger values to represent greater probabilities of target occurrence.

REAL DATA RESULTS

We illustrate the effectiveness of the multiphysics anomaly map by showing examples of the application in two real data examples from Brazil: onshore Recôncavo Basin and offshore Sergipe-Alagoas.

Recôncavo Basin example

The first example comes from the onshore Recôncavo Basin, NE of Brazil, where the identification of thick hydrocarbon-bearing sand bodies is not straightforward in the seismic data alone. Five wells named REC-1, REC-2, REC-3, REC-4 and REC-5 were drilled in the area with only one discovery (REC-1). Because the reservoir thickness in these wells were respectively 43 m, 39 m, 28 m, 16 m, and 10 m, the interpreters concluded that the chance of finding a commercial accumulation was directly related to the thickness of the sandstones. The low quality of the seismic data in the area combined with the great variability of the sandstones make the correlation between seismic attributes and sandstone thickness not always straightforward.

To address that issue Ramos et al. (2011) combined the seismic amplitude (Figure 2a) with a density contrast distribution (Figure 2b) derived from a 3D gravity inversion in the first approach to the MAM tool. They verified, via density and velocity logs information, that the sand bodies would generate both strong seismic amplitude and negative density contrast anomalies. A new well (REC-6) drilled based on their results have found a new hydrocarbon-bearing sandstone with the largest thickness among all other wells in the area.

In this paper, we extend the work of Ramos et al. (2011) to illustrate the effectiveness of the multiphysics anomaly map. We selected the same seismic amplitude (Figure 2a) and density anomalies (Figure 2b) maps to keep consistency with previous work.

The seismic amplitude anomaly domain is (0.15,5.10) and shows a right-skewed distribution with a mean of 1.3, and standard deviation of 0.5, as shown in the box on the right upper corner of Figure 2a. Such distribution lead us to take the domain (0.15,2.26) as the 95% interval. The data was then shifted by -1.205 to place the domain center in zero, and the scale parameter was defined to be 22.75 so that the final domain became (-24,24). A logistic function with parameters k=0.25, L=2, s=22.75 and x_0=0.545 was then applied to the seismic amplitude data to transform it in the seismic logistic anomaly shown in Figure 3a.

Because the sandstones show negative density contrast, it is necessary to change the sign of the density anomaly so that the negative anomalies become more significant for the logistic function. The data domain for the density anomaly was (-2.8,4.9) and its distribution is left-skewed with mean close to zero and standard deviation equals to 1.3, as seen in the box on the right upper corner of Figure 2b. The 95% interval for the density anomaly is then (-2.56,2.56) and no shift is necessary since the mean is already close to zero. The density data is scaled by 9.375 to adjust the interval to (-24,24). The parameters k=0.25, L=2 and x_0=0.5 were used to define the logistic function that transformed the density data into a density logistic anomaly (Figure 3b).

The logistic anomalies were multiplied together to form the multiphysics anomaly map of Figure 3c. Because the lateral continuity of the sandstones is not evident in seismic data due to the discontinuity of the reflections, the combination with density contrast anomalies unveil a remarkable anomaly resembling a channel that crosses the area in W-E direction. This is in agreement with classical exploratory models where the location of large sandstone thickness is conditioned by troughs located along the boundary of shale diapirs (areas with no data in figures 2a, 3a and 3c). The REC-6 well, located on the top of this anomaly was drilled based on Ramos et al. (2011) results and have found the largest sandstone thickness among all wells in the area. On the other hand, REC-5 well drilled away from any anomaly is the well with the smallest sandstone thickness. The present result makes clear the value of the proposed strategy as an exploration tool.

Sergipe-Alagoas Basin example

The second example comes from offshore Sergipe-Alagoas Basin, NE of Brazil. Although seismic interpretation could successfully identify two channels, drilling has confirmed that just one of them was hydrocarbon bearing. Analysis of seismic attributes alone proved to be ambiguous. To reduce the ambiguity in the interpretation, we used a resistivity anomaly map derived from a 3D unconstrained inversion of a Control Source Electromagnetic (CSEM) data acquired in the area (Lorenz et al., 2013). Because the post-salt oil-bearing reservoirs at the Brazilian offshore frequently exhibit a resistive anomaly, the CSEM method can be used for drilling decision (Buonora et al., 2014) since it can differentiate from resistive (oil-bearing) and non-resistive (water-bearing) reservoirs.

For the MAM construction we used two band-limited seismic inversion attributes (Waters and Kemper, 2014), VP/VS (Figure 4a) and IP-IS (Figure 4b), both providing anomalous channel forms (labeled **A** and **B** in Figures 4a and 4b) and the resistivity data (Figure 4c). Considering that anomalous band–limited seismic inversion attributes in this case are represented by negative values, it was necessary to change the signs of both attributes.

The band-limited VP/VS domain is (-0.28,0.36) and its distribution is close to symmetric with mean equals to 0.02 and standard deviation equals to 0.06, as shown in the box on the right upper corner of Figure 4a. The 95% domain interval becomes (-0.10,0.14) and was then shifted to (-0.12,0.12) to place the center in zero. The VP/VS data were scaled by 200 to adapt to the (-24,24) interval. A logistic function with parameters k=0.25, L=2 and x_0=0.03 was applied to the data transforming it into the VP/VS logistic anomaly of Figure 5a.

The data domain for the band-limited IP-IS attribute is (-518, 838) showing also a symmetric distribution with mean equals to 13.7 and standard deviation equals to 118.6, as seen in the box on the right upper corner of Figure 4b. The interval (-225, 251) was selected as the 95% interval for this dataset. The data was then shifted to (-238,238) and scaled by 0.1 to fit in the (-24,24) interval. The parameters k=0.25, L=2 and x_0=137 were chosen for the logistic function which transform the data into an IP-IS logistic anomaly of Figure 5b.

The resistivity data domain exhibits a right-skewed distribution with mean equals to 26.3 and standard deviation equals to 18.2 in the interval (2.6,121.6) and therefore, its 95% becomes (2.6,63.0) as seen in the box on the right upper corner of Figure 4c. The domain is shifted to (-30.2,30.2) to place the center in zero and scaled by 0.8 to adapt to the interval (-24,24). We have selected a logistic function with parameters k=0.25, L=2 and x_0=-2.8 to transform the IP-IS dataset into the resistivity logistic anomaly of Figure 5c.

The logistic anomalies resistivity, band-limited VP/VS, and IP-IS were multiplied to create the multiphysics anomaly map of Figure 5d. The resulting MAM map shows a strong anomaly over the oil-bearing channel (labeled **A**) as demonstrated by three successful drilled wells in the area (wells 1, 2, and 5 in Figure 5d). On the other hand, MAM shows a faded anomaly, in agreement with the geological knowledge of two dry well drilled on channel B (wells 3 and 4 in Figure 5d).

Because the shape of the resistivity anomaly does not resemble the channel shape seen in the seismic anomalies, just the visual inspection of the three maps is not enough to provide any insight about the channel's content. After combining the maps into the multiphysics anomaly becomes clear that there is a difference between the channels, being channel A more resistive than channel B. The difference in resistivity can be interpreted as presence of oil in channel A since oil is more resistive than water. In this case, the multiphysics characteristics of MAM has proved to be a risk-reducing tool, helping in reducing the ambiguity of the traditional seismic interpretation.

CONCLUSION

We have developed a new strategy for integrating multi-sensor geophysical data into a single dataset specially designed to emphasize some qualitative aspects of the geological interpretation. The proposed methodology extracts the most important anomalies of each input dataset and highlights regions of high correlation, which are therefore interpreted as significant geological features. The strength in the use of MAM lies in its ability to synthesize multiple interpretation providing a view of the data that is not possible to achieve with individual geophysical interpretation. The results described in the real data examples make clear the power and flexibility of the multiphysics anomaly map for different geological problems, each of them requiring a specific combination of geophysical methods to produce consistent results.

Because the multiphysics anomaly map is strongly problem dependent, the choice of input geophysical data is of fundamental importance to the success of the methodology. As the anomalies chosen to compose the multiphysics anomaly map need to be related to the same geological source, it is necessary to identify and isolate anomalies, which is not a simple task and becomes therefore, a limitation for the construction of the multiphysics anomaly map. Considering that the multiphysics anomaly map combines different physical units it is important to reinforce its qualitative nature, where the relative relationships between anomalies are more significant than their values themselves. In the future, it seems straightforward to incorporate geological elements into the process. Structural and stratigraphic features, for instance, can be integrated in the multiphysics anomaly map to visually inspect their spatial relationships with the geophysical anomalies, demonstrating the usefulness of our method as an interpretation tool.

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in the box on the right upper corner of (c).

FIGURE 5: Logistic anomalies and the resulting MAM for Sergipe-Alagoas Basin example. (a) The logistic band-limited VP/VS anomaly calculated with threshold value equals to 0.03. (b) The logistic band-limited IP-IS anomaly was computed using threshold value x_0=137. (c) The logistic resistivity anomaly with threshold x_0=-2.8. SEAL-1, SEAL-2 and SEAL-5 are producing wells. SEAL-3 and SEAL-4 are dry wells. (d) The computed MAM shows strong amplitudes in the oil-bearing channel **A**.

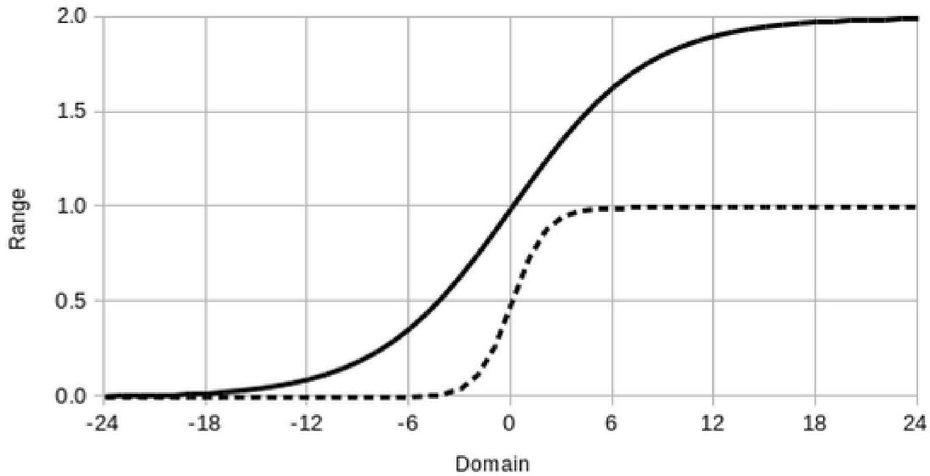

FIGURE 1: Logistic functions plotted in domain (-24,24). The standard sigmoid function (range L=1, steepness k=1) represented by the dashed line is a special case of the generalized logistic function, while the solid line represents the modified logistic function (L=2, k=0.25) proposed in this work. Notice that for the modified logistic function, x-values greater than the threshold value x0 =0 will be enhanced while smaller ones will be reduced. On the other hand, for the standard sigmoid function, all x-values will be reduced since the function output equals 1 only at x=+∞.

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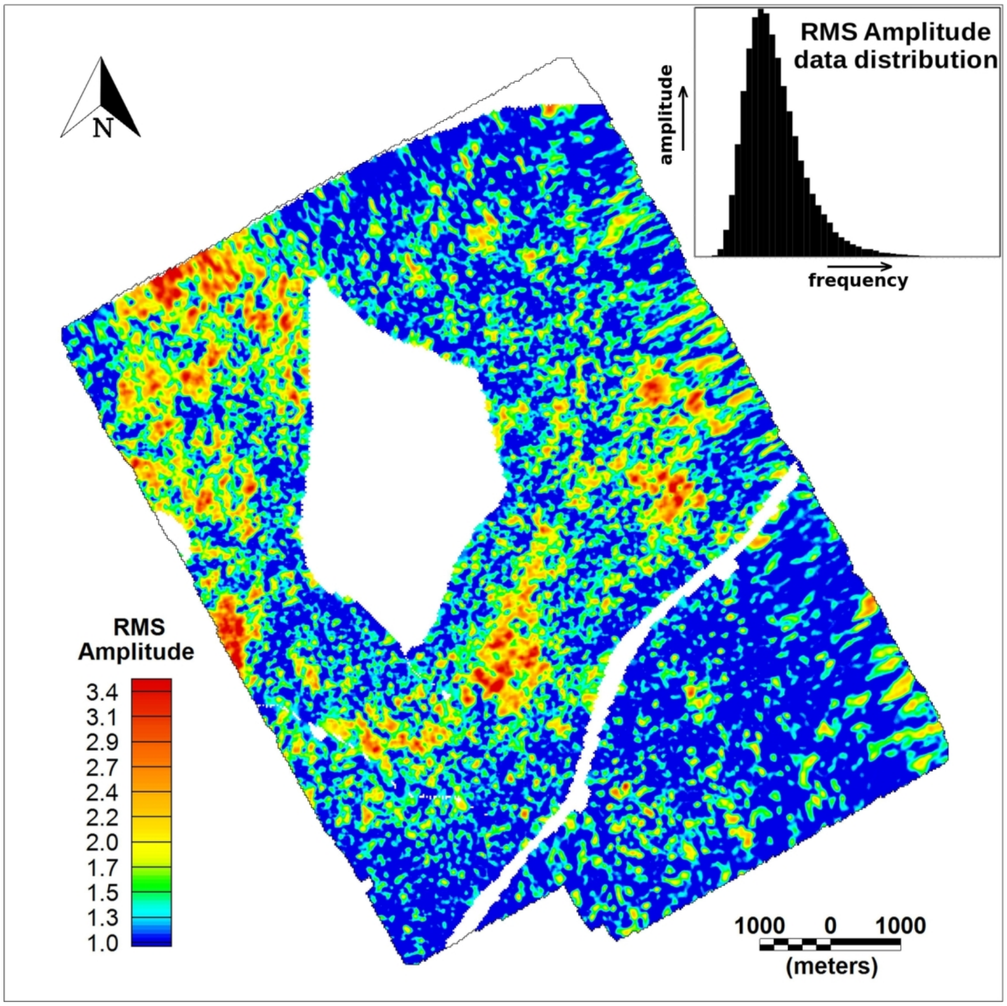

FIGURE 2a: Anomalies selected for multiphysics anomaly map computation in Recôncavo Basin and their respective data distributions. The RMS amplitude distribution from seismic interpretation in (a) exhibits a clearly right-skewed distribution as shown in the box on the right upper corner of (a). Areas with no data in (a) result from absence of the mapped seismic horizon. The density contrast distribution derived from a residual gravity anomaly in (b) shows a left-skewed distribution as seen in the box on the right upper corner of (b).

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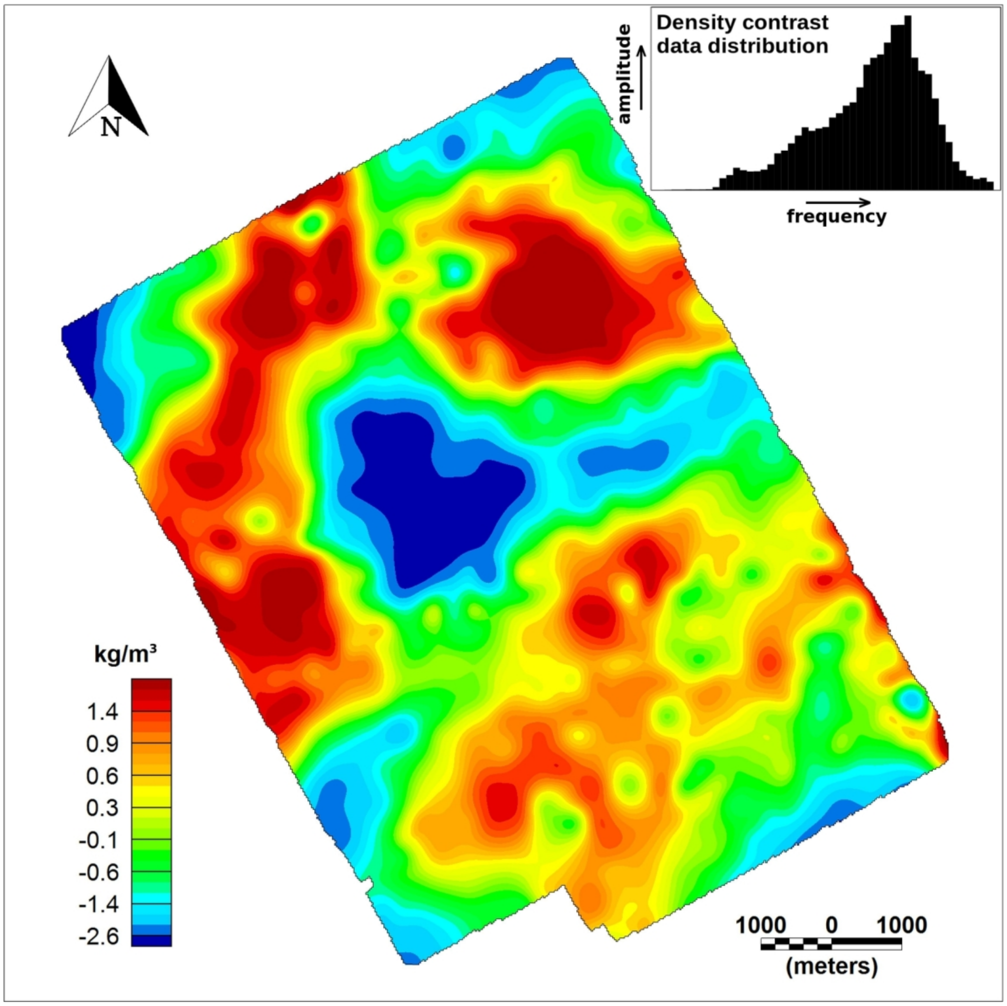

FIGURE 2b: Anomalies selected for multiphysics anomaly map computation in Recôncavo Basin and their respective data distributions. The RMS amplitude distribution from seismic interpretation in (a) exhibits a clearly right-skewed distribution as shown in the box on the right upper corner of (a). Areas with no data in (a) result from absence of the mapped seismic horizon. The density contrast distribution derived from a residual gravity anomaly in (b) shows a left-skewed distribution as seen in the box on the right upper corner of (b).

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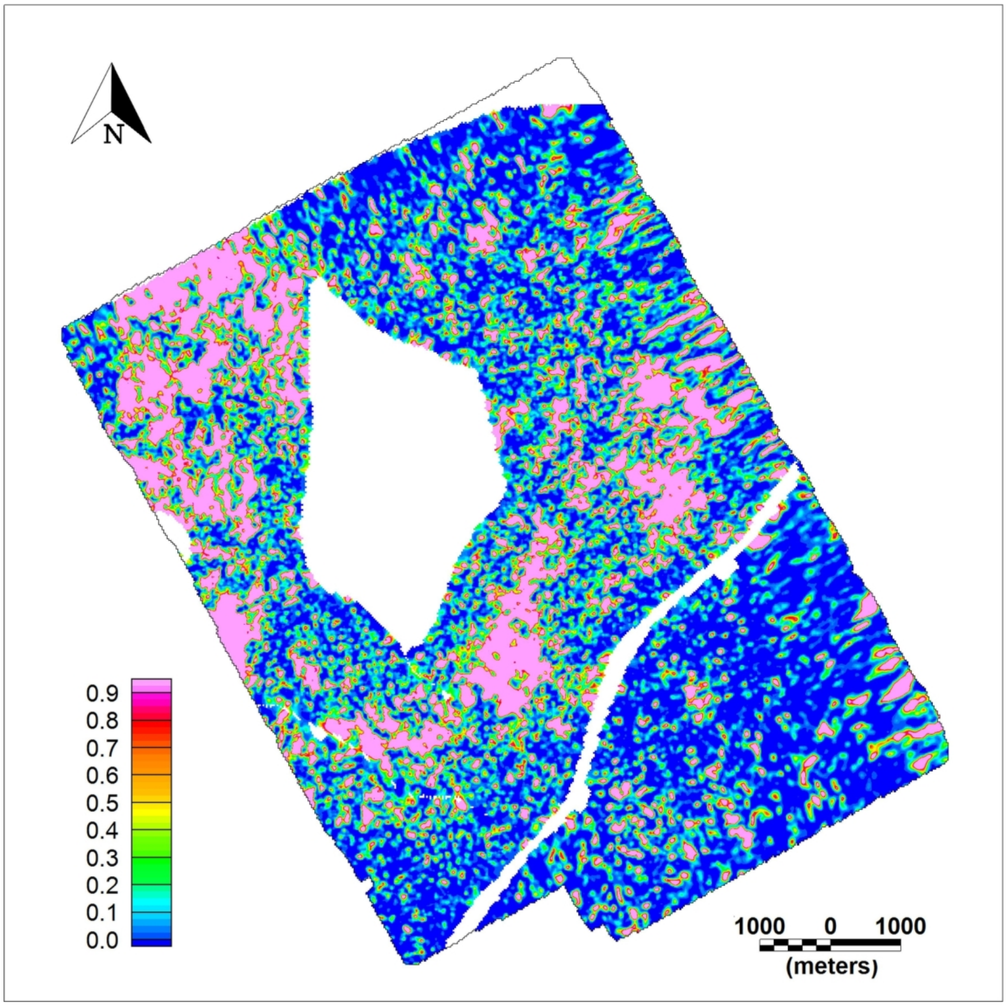

FIGURE 3a: Logistic anomalies and the resulting MAM for Recôncavo Basin example. (a) Seismic logistic anomaly considering threshold value x0 =0.545. (b) Density logistic anomaly computed with threshold value x0 = 0.5. (c) multiphysics anomaly map resulting from the multiplication of the logistic functions (a) and (b). Notice that the region with largest values, interpreted as favorable to the occurrence of targets, exhibits a typical channel shape. While REC-5 well has drilled just a small amount of sandstone, REC-6 well have found the largest sandstone thickness among all the wells in the area (Ramos et al., 2011).

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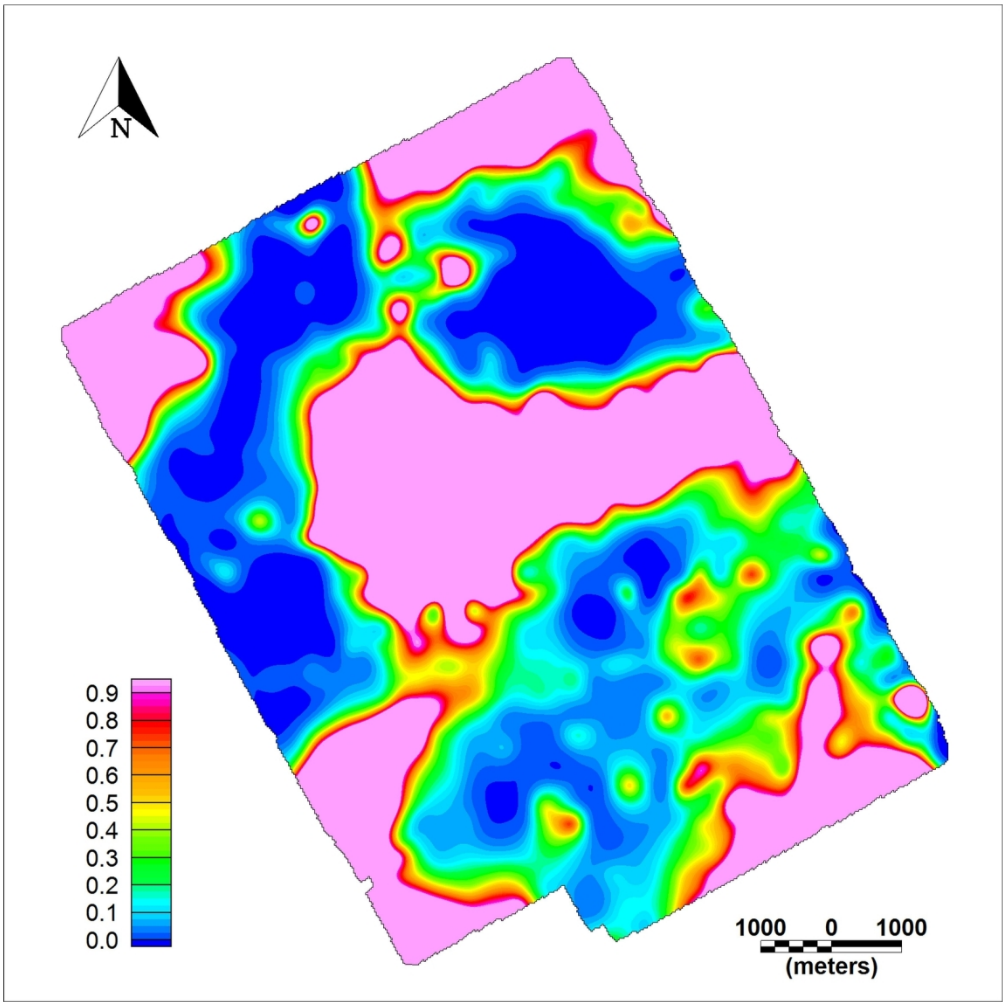

FIGURE 3b: Logistic anomalies and the resulting MAM for Recôncavo Basin example. (a) Seismic logistic anomaly considering threshold value x0 =0.545. (b) Density logistic anomaly computed with threshold value x0 = 0.5. (c) multiphysics anomaly map resulting from the multiplication of the logistic functions (a) and (b). Notice that the region with largest values, interpreted as favorable to the occurrence of targets, exhibits a typical channel shape. While REC-5 well has drilled just a small amount of sandstone, REC-6 well have found the largest sandstone thickness among all the wells in the area (Ramos et al., 2011).

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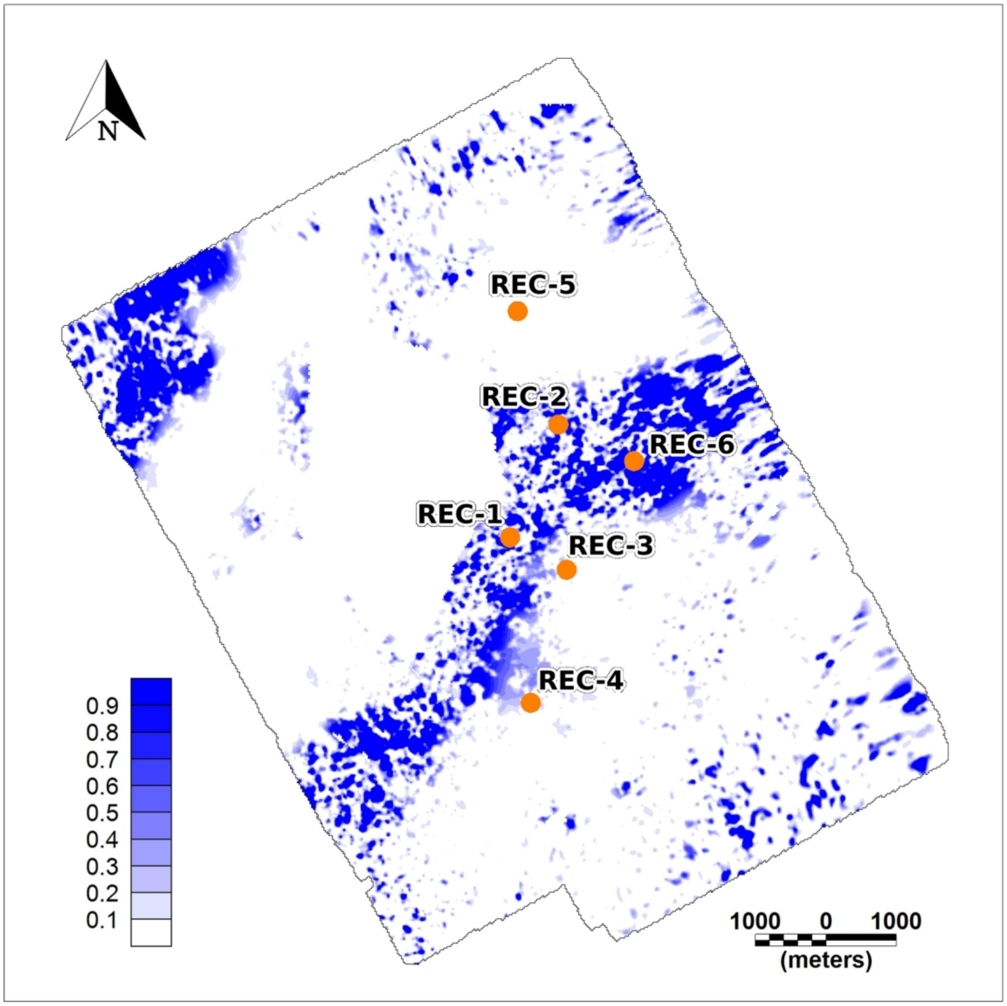

FIGURE 3c: Logistic anomalies and the resulting MAM for Recôncavo Basin example. (a) Seismic logistic anomaly considering threshold value x0 =0.545. (b) Density logistic anomaly computed with threshold value x0 = 0.5. (c) multiphysics anomaly map resulting from the multiplication of the logistic functions (a) and (b). Notice that the region with largest values, interpreted as favorable to the occurrence of targets, exhibits a typical channel shape. While REC-5 well has drilled just a small amount of sandstone, REC-6 well have found the largest sandstone thickness among all the wells in the area (Ramos et al., 2011).

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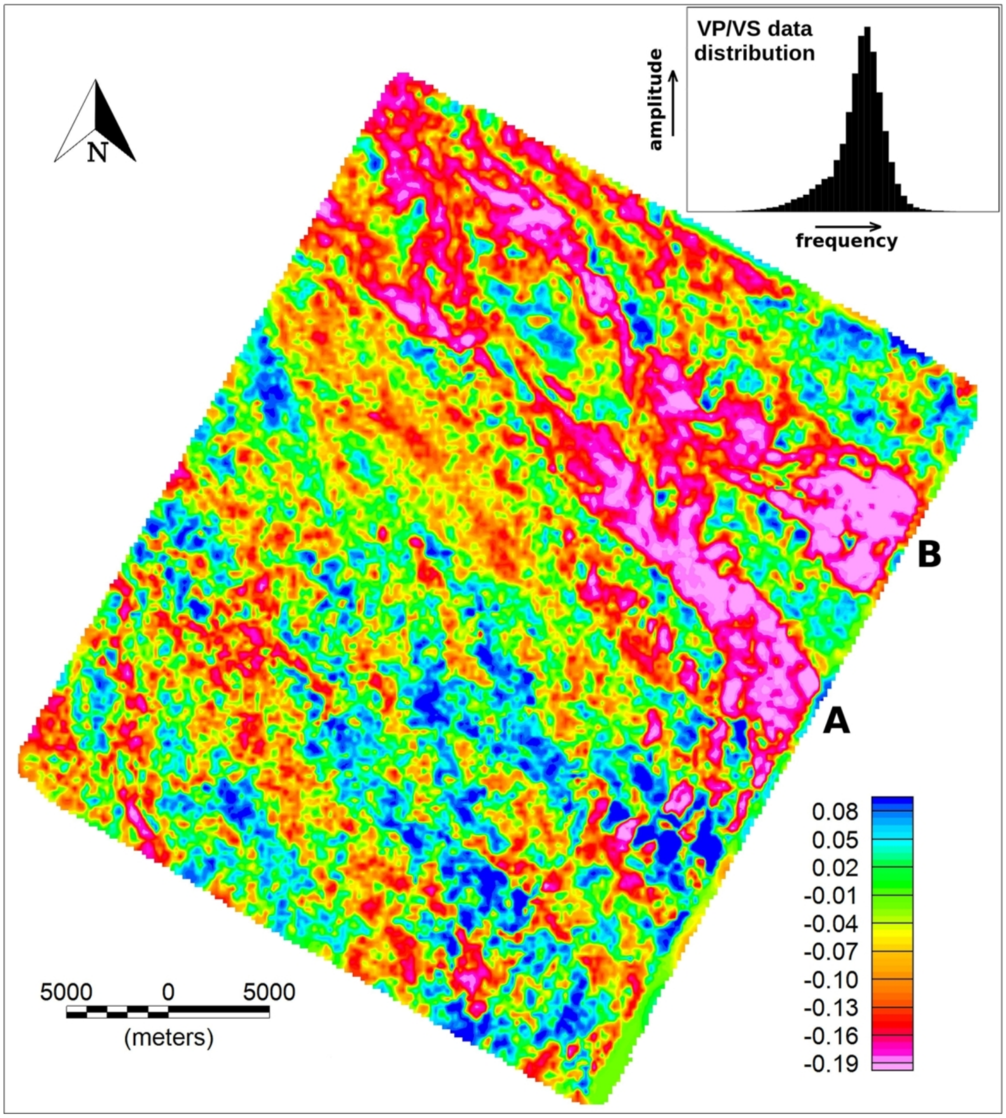

FIGURE 4a: Anomalies selected for MAM construction in Sergipe-Alagoas Basin and their respective data distribution. Amplitude maps extracted along a seismic horizon for band-limited VP-VS attribute in (a) and band-limited IP-IS attribute in (b) show an approximately symmetric distributions as illustrated in the boxes on the right upper corner of (a) and (b), respectively. On the other hand, the resistivity anomaly map in (c) exhibits a typical right-skewed distribution as shown in the box on the right upper corner of (c).

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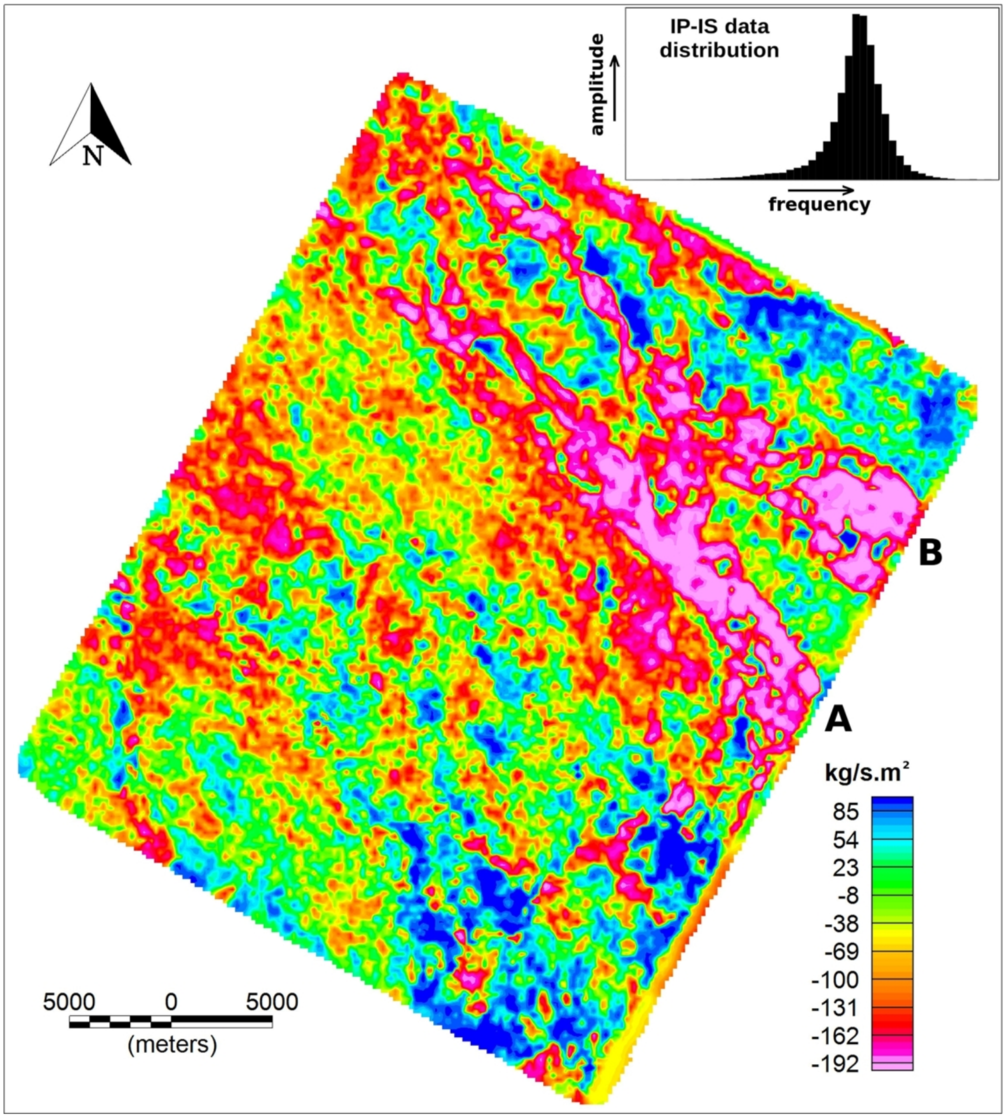

FIGURE 4b: Anomalies selected for MAM construction in Sergipe-Alagoas Basin and their respective data distribution. Amplitude maps extracted along a seismic horizon for band-limited VP-VS attribute in (a) and band-limited IP-IS attribute in (b) show an approximately symmetric distributions as illustrated in the boxes on the right upper corner of (a) and (b), respectively. On the other hand, the resistivity anomaly map in (c) exhibits a typical right-skewed distribution as shown in the box on the right upper corner of (c).

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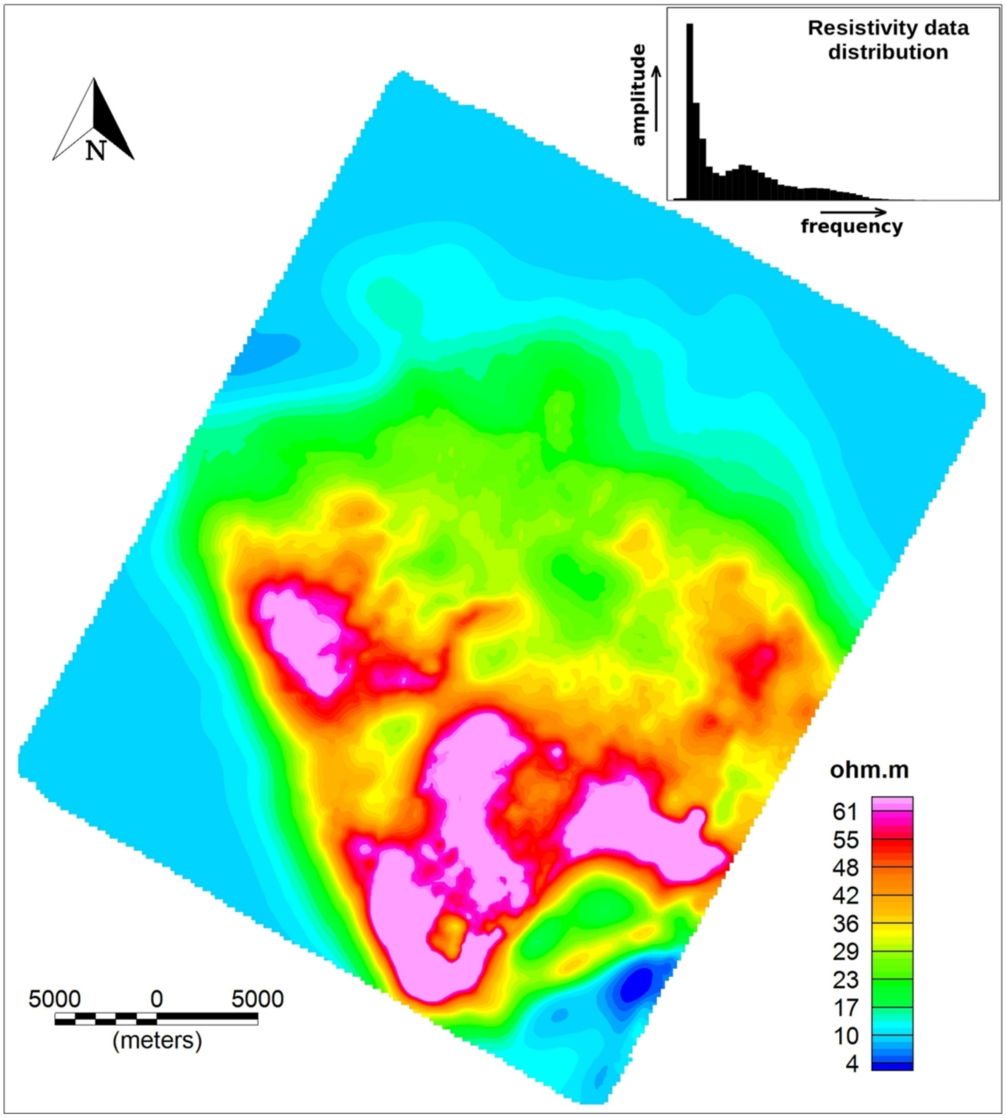

FIGURE 4c: Anomalies selected for MAM construction in Sergipe-Alagoas Basin and their respective data distribution. Amplitude maps extracted along a seismic horizon for band-limited VP-VS attribute in (a) and band-limited IP-IS attribute in (b) show an approximately symmetric distributions as illustrated in the boxes on the right upper corner of (a) and (b), respectively. On the other hand, the resistivity anomaly map in (c) exhibits a typical right-skewed distribution as shown in the box on the right upper corner of (c).

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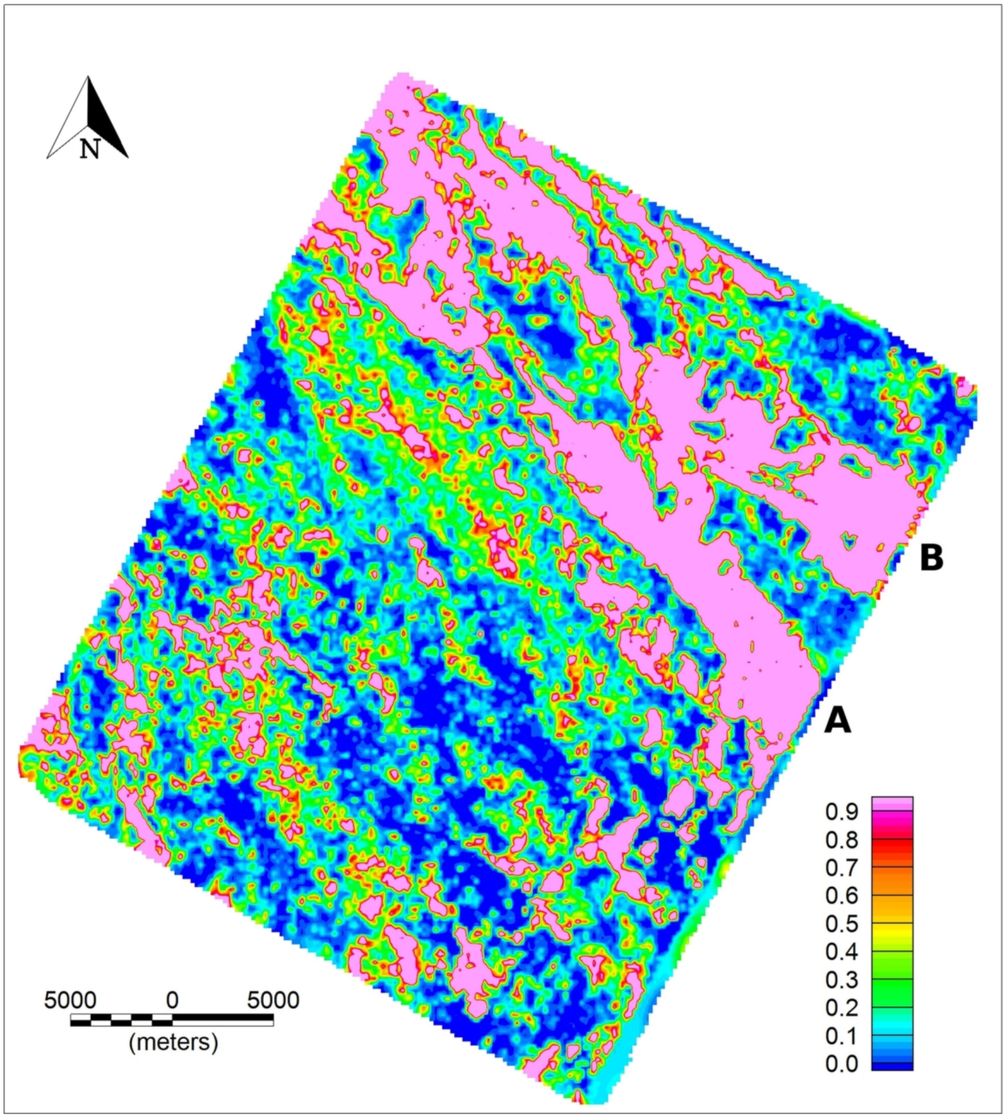

FIGURE 5a: Logistic anomalies and the resulting MAM for Sergipe-Alagoas Basin example. (a) The logistic band-limited VP/VS anomaly calculated with threshold value equals to 0.03. (b) The logistic band-limited IP-IS anomaly was computed using threshold value x0 =137. (c) The logistic resistivity anomaly with threshold x0 =-2.8. SEAL-1, SEAL-2 and SEAL-5 are producing wells. SEAL-3 and SEAL-4 are dry wells. (d) The computed MAM shows strong amplitudes in the oil-bearing channel A.

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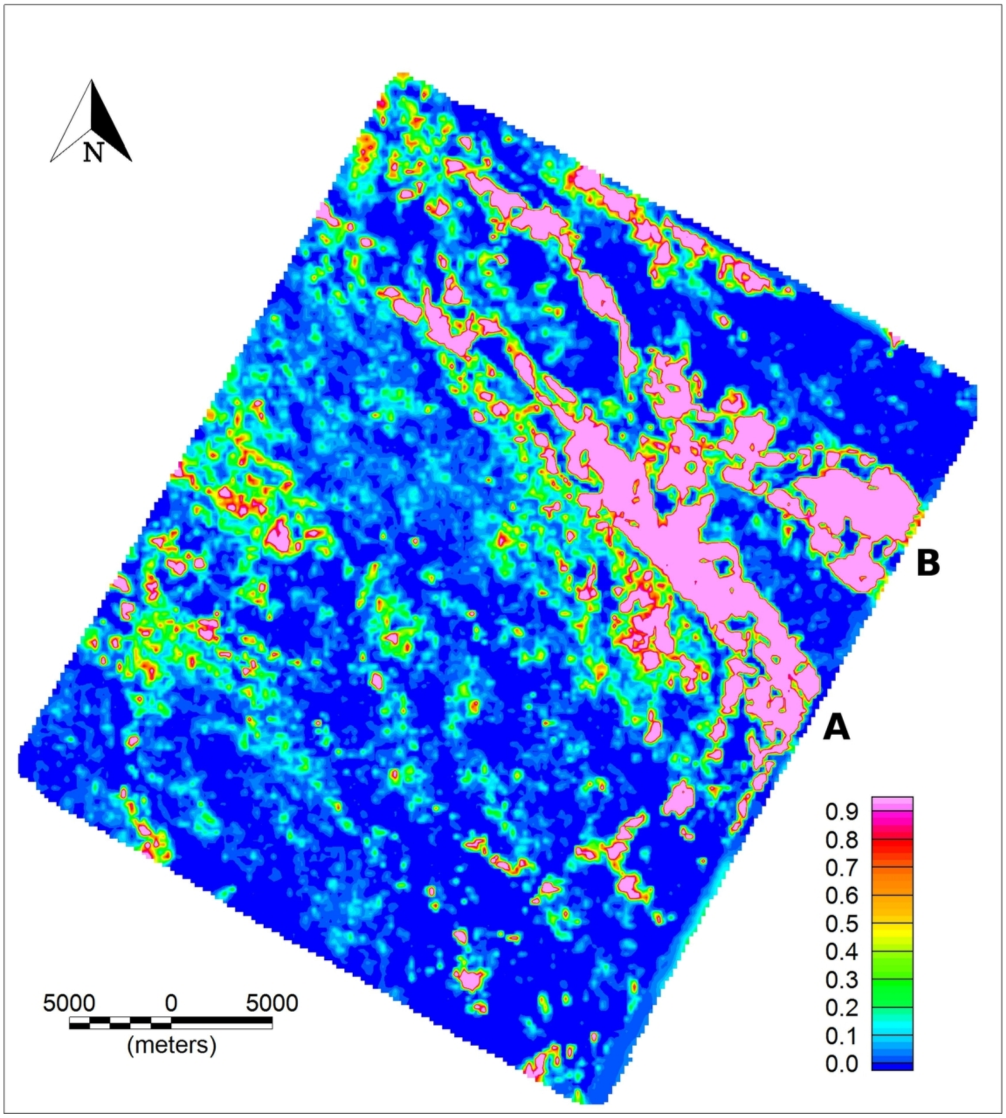

FIGURE 5b: Logistic anomalies and the resulting MAM for Sergipe-Alagoas Basin example. (a) The logistic band-limited VP/VS anomaly calculated with threshold value equals to 0.03. (b) The logistic band-limited IP-IS anomaly was computed using threshold value x0 =137. (c) The logistic resistivity anomaly with threshold x0 =-2.8. SEAL-1, SEAL-2 and SEAL-5 are producing wells. SEAL-3 and SEAL-4 are dry wells. (d) The computed MAM shows strong amplitudes in the oil-bearing channel A.

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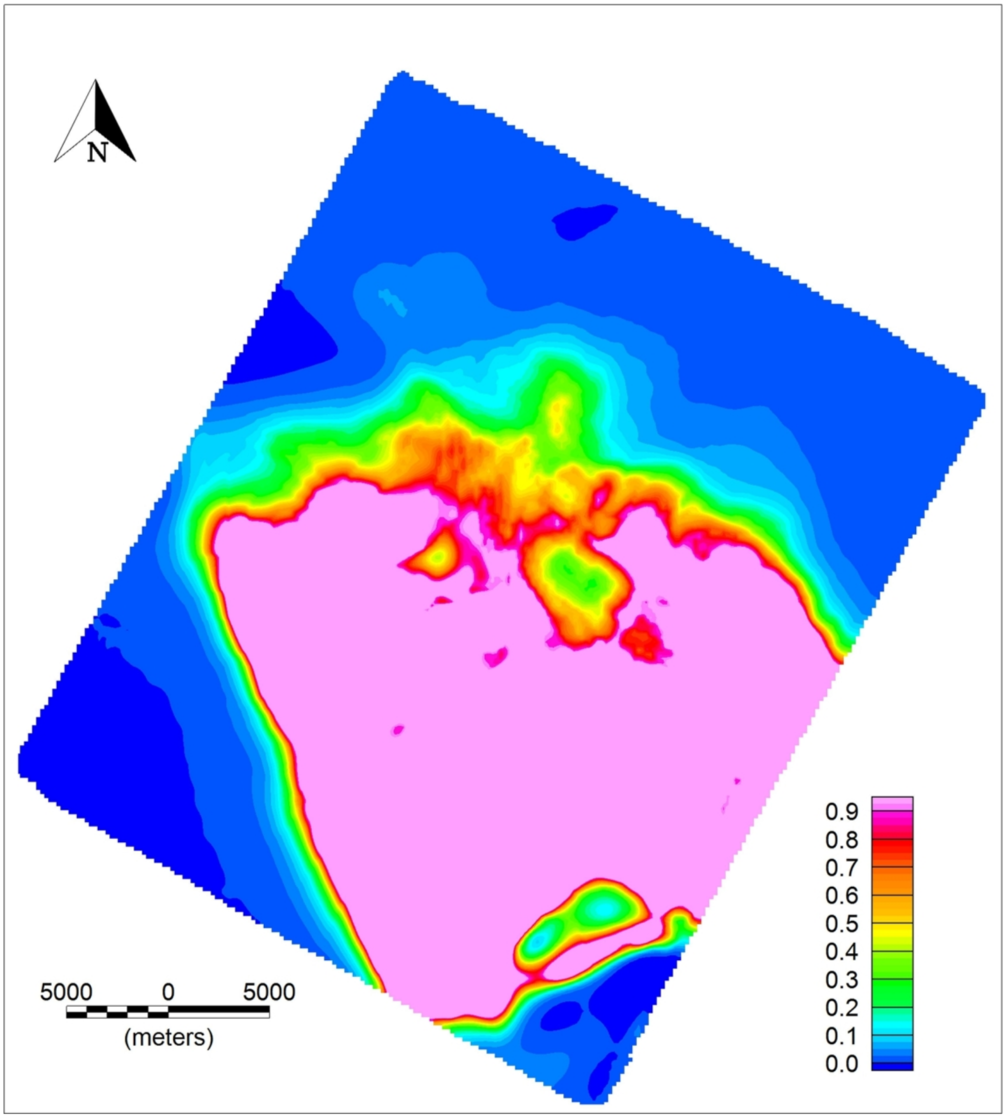

FIGURE 5c: Logistic anomalies and the resulting MAM for Sergipe-Alagoas Basin example. (a) The logistic band-limited VP/VS anomaly calculated with threshold value equals to 0.03. (b) The logistic band-limited IP-IS anomaly was computed using threshold value x0 =137. (c) The logistic resistivity anomaly with threshold x0 =-2.8. SEAL-1, SEAL-2 and SEAL-5 are producing wells. SEAL-3 and SEAL-4 are dry wells. (d) The computed MAM shows strong amplitudes in the oil-bearing channel A.

146x162mm (300 x 300 DPI)

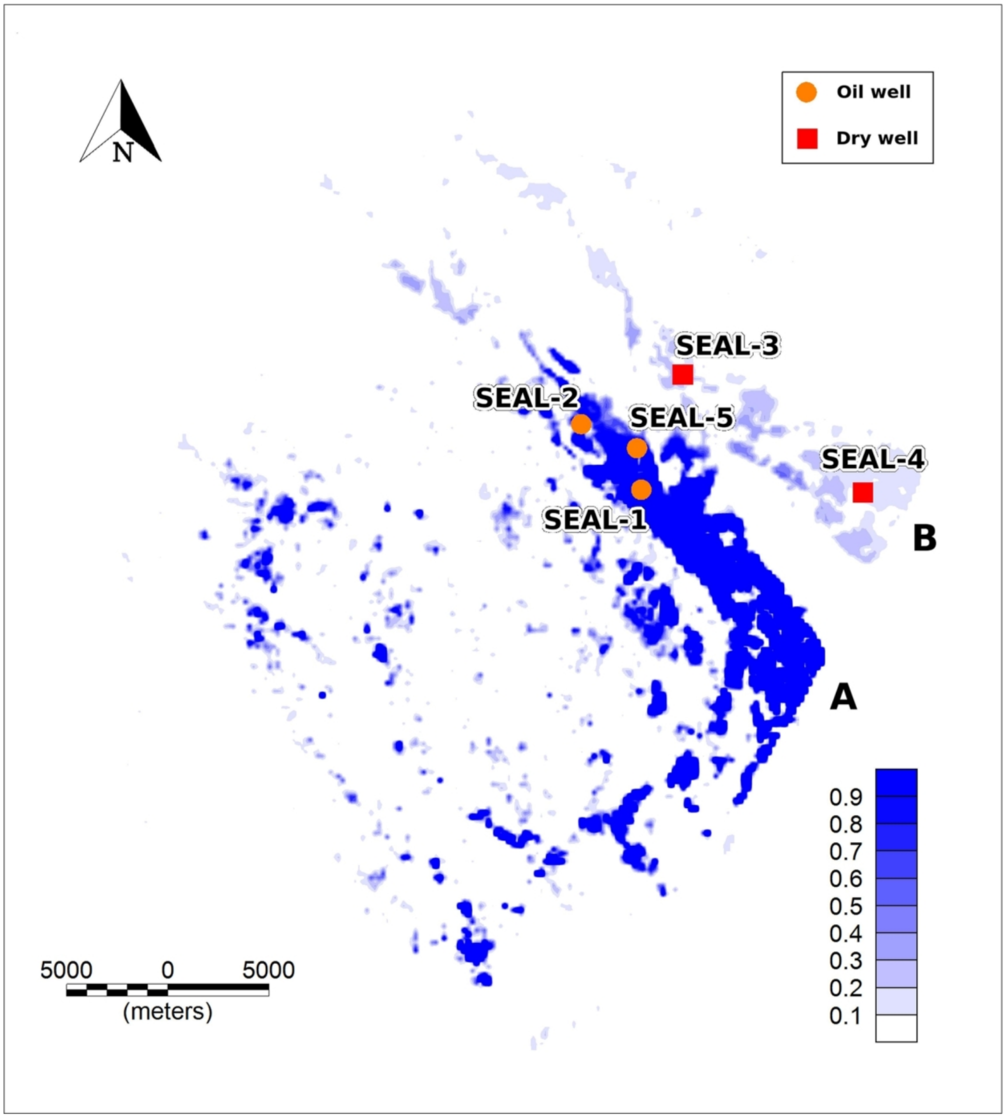

FIGURE 5d: Logistic anomalies and the resulting MAM for Sergipe-Alagoas Basin example. (a) The logistic band-limited VP/VS anomaly calculated with threshold value equals to 0.03. (b) The logistic band-limited IP-IS anomaly was computed using threshold value x0 =137. (c) The logistic resistivity anomaly with threshold x0 =-2.8. SEAL-1, SEAL-2 and SEAL-5 are producing wells. SEAL-3 and SEAL-4 are dry wells. (d) The computed MAM shows strong amplitudes in the oil-bearing channel A.

146x162mm (300 x 300 DPI)

DATA AND MATERIALS AVAILABILITY

Data associated with this research are confidential and cannot be released.